

Softcom

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CNN

→ দুটি ইন্টি pattern খুঁজতে হবে,

$$\text{filter} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

→ filter জিলায় দেবে ২টি pattern
জিলা ফিল্টার,

$$\text{identify} = \begin{bmatrix} X \\ 0 \end{bmatrix}$$

dot product বো ফিল্টার স্ক্রল,

convolution = if a f^n multiplies a f^n and
return another f^n .

convolution layer ১ padding use করা হয়,

Left
diagonal

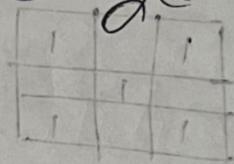
x

Right
diagonal

→ 3 kernel

→ padding add କେବଳ size same ଅଟେ,

→ ଏହାକୁ filter ପଦ୍ଧତିରେ ବୁଝା image.



pooling

→ shrink the image.

→ pooling window traverse like filter

→ Max pooling → get most dominant pixel.

→ Avg pooling → each pixel ko contribution

→ square matrix

→ stride → କେଉଁ କେଉଁ ଅଟେ,

→

stride निम्न सूत्रों से देता है,

egⁿ:

$$\text{floor} \left[\frac{W/H - f + 2 * p}{S} \right] + 1$$

$S = \text{stride}$ | $W = \text{width}$

$f = \text{filter}$

$H = \text{Height}$

$p = \text{padding}$

Say,

$$W = 28, H = 28$$

$$S = 2, p = 0, f = 2$$

$$\text{floor} \left[\frac{28 - 2 + 2 * 0}{2} \right] + 1$$

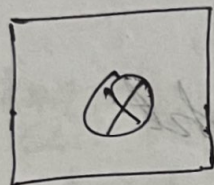
$$= 14$$

⇒ CNN feature size ko kam karke feature

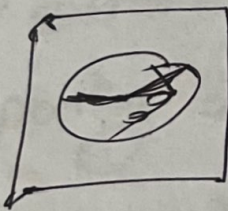
size kam karke,

⇒ Max pooling unnecessary feature ko kam karke

feature



⇒ convolution



⇒

⇒ Neg value ko 0 karke diya,

ReLU → positive → value ko
negative → 0

CNN normalization.

⇒ Mean pooling use karke neg value ko kam karke
value ko kam karke ReLU

1 layer

convolution $\rightarrow$ RELU $\rightarrow$ pooling

standard procedure

$\rightarrow$ RELU - negative values convert to zero

$\rightarrow$ dropout $\rightarrow$ randomly drop pixel & neurons

also

$\rightarrow$ model training is biased

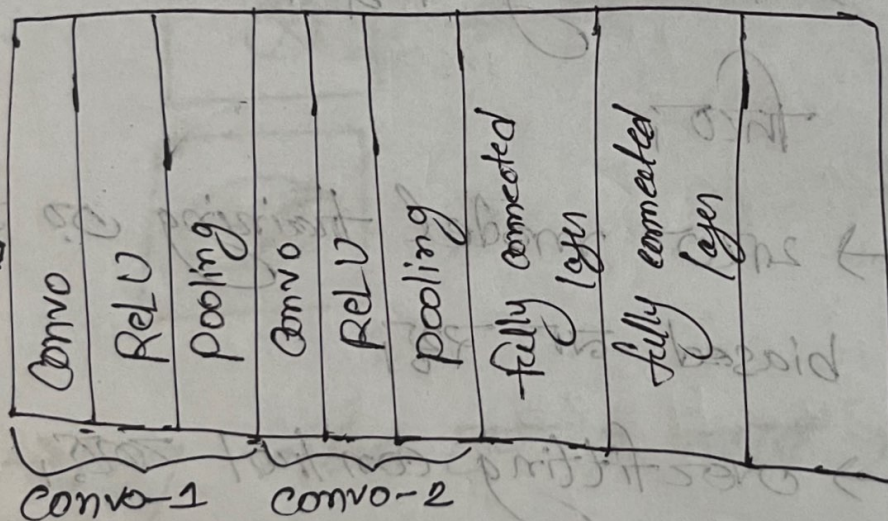
$\rightarrow$ overfitting control

$\rightarrow$ Multiple conv layers use

$\rightarrow$ each filter is a feature

→ convolution 50 तक ब्रॉड डीप प्रिपैरे करें

→ CNN एक feature extractor



⇒ 1D में convolution 50 feature 50 तक
होना चाहिए,

⇒ 2D या 3D लायर्स,

→ data 3 column change કરાવે પ્રોબ

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→ column ઉમેરે જાનિતે કરાવે એક data
same wastell યાદે ઉછાતે conv use
કરે યાદે ના,

Model Architecture :

conv

kernel = 5

stride = 1

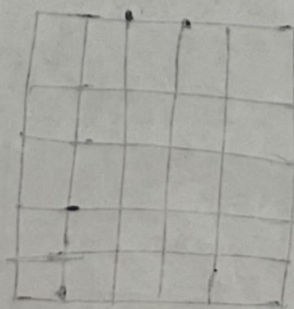
$P = 2$

pooling

$K = 2$

$S = 2$

$P = 0$



input ની રીડે
2x2 ની રીડે